

Work Report: Computer Vision Project Using YOLOv8

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Project Overview

This project focused on developing and evaluating object detection models using the YOLOv8 framework. The work was divided into two primary phases: training a model on a pre-built dataset from Roboflow and creating a custom dataset for further training. The objective was to gain practical experience in dataset preparation, model training, evaluation, and deployment, with models exported to ONNX format for cross-platform compatibility.

Phase 1: Training on Roboflow Dataset

Dataset and Tools

- Source: Utilized a pre-built dataset from Roboflow, which included annotated images and a data.yaml configuration file.
- Framework: Employed the Ultralytics YOLOv8 library for model training and evaluation.

Methodology

- Model Training: Trained a YOLOv8n model for 3 epochs using the dataset located at C:/Users/gunem/OneDrive/Desktop/let's move/rakshak work/data/data.yaml.
- Evaluation: Calculated performance metrics, specifically mean Average Precision (mAP) at IoU thresholds of 0.5 (mAP@0.5) and 0.5 to 0.95 (mAP@0.5:0.95).
- Export: Converted the trained model to ONNX format with dynamic input shapes for deployment flexibility.

Results

- mAP@0.5: 0.148
- mAP@0.5:0.95: 0.052
- Export: Successfully exported to best.onnx

Phase 2: Training on Custom Dataset

Dataset Creation

- Data Collection: Gathered approximately 120 images from Google and other online sources.
- Annotation: Used CVAT to annotate images with bounding boxes, saving annotations in YOLO format.
- Organization: Structured the dataset into images/train and labels/train folders, with a config.yaml file specifying paths and class details (1 class: 'target').

Methodology

- Model Training: Trained a YOLOv8n model for 5 epochs using the custom dataset at C:/Users/gunem/OneDrive/Desktop/data/config.yaml.
- Evaluation: Assessed model performance using mAP@0.5 and mAP@0.5:0.95 metrics.
- Export: Exported the trained model to ONNX format.

Results

- mAP@0.5: 0.004
- mAP@0.5:0.95: 0.0015
- Export: Successfully exported to best.onnx

Challenges and Observations

- The Roboflow dataset yielded better mAP scores, likely due to its larger size (644 training images vs. 119 in the custom dataset) and higher-quality annotations.
- The custom dataset model showed lower performance, possibly due to limited data and potential inconsistencies in manual annotations.

Learning Outcomes

Through this project, I developed hands-on expertise in:

- Dataset preparation and annotation using CVAT.
- Object detection workflows with YOLOv8.
- Model evaluation using mAP and IoU metrics.
- Model export to ONNX for deployment.

Additionally, I enhanced my understanding of mAP and IoU concepts through YouTube tutorials and practical application.

Conclusion

This project successfully demonstrated the end-to-end process of training and deploying YOLOv8 models for object detection. While the Roboflow dataset provided a robust starting point, creating a custom dataset offered valuable insights into data collection and annotation challenges. Future improvements could include expanding the custom dataset.